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Research Proposal

on

Certainty Without Accuracy: Effects of Generative-AI Confidence Expression on Delayed Unaided Circuit-Fault Diagnosis

Scholar Name

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Islamic University of Technology
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Date of Submission

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Introduction:

Generative artificial intelligence (AI) is currently widely used in engineering education. A study conducted in 2026 on 378 engineering students from Bangladesh revealed that behavioural intention was the main factor predicting the use of generative AI, the model accounting for 64% of the variation in users' behaviour [1]. Nevertheless, the impact of generative AI on students' capacity to make technical decisions on their own is still uncertain.

This risk is particularly important in electric circuits. In a study of 72 undergraduate circuit problems, ChatGPT-4o struggled to interpret some circuit configurations correctly [2]. When it made a mistake in understanding the circuit, that mistake often affected the calculations and the final answer. This could be a problem for students who may not yet have enough knowledge to recognize the error.

Correctness and confidence are independent properties of the output, and in this case only one is visible to the student. Moreover, people often use an advisor's expressed confidence as an indicator of the advice's reliability. Research on advice-taking has shown that more confident recommendations are more likely to be accepted than less confident ones, although confidence does not necessarily correspond to actual accuracy [3]. With a human advisor, confidence reflects their personal mindset, thoughts and conviction. But with a generative AI model, it is a phrasing choice; adjustable by whoever writes the system prompt.

Two related bodies of research are relevant here: how an AI's expressed certainty affects immediate reliance on its advice [4], [5], and how AI-assisted practice affects performance after the AI is removed [6]. These areas have not yet been directly connected.

In the field of electrical engineering, this issue has practical significance. If an AI or software program gives an incorrect answer, engineers have to be capable of analyzing the circuit themselves and spotting the error. It follows that the ability to diagnose faults is a fundamental skill which must still work effectively even without the aid of AI. The present study looks at whether the level of certainty displayed by the AI during practical work is connected to students' later ability to independently diagnose faults in circuits.

2. Literature Review

Advice-taking. Research on advice-taking shows that decision-makers often retain much of their initial judgment rather than fully accept an advisor's recommendation. Bonaccio and Dalal describe this pattern as *egocentric advice discounting* and found that decision-makers typically shift only about 20–30% toward an advisor's recommendation [3]. However, an advisor's expressed confidence can influence how much weight is given to the advice, even when confidence does not necessarily reflect accuracy [3]. This provides a basis for examining whether expressed confidence in AI advice has a similar effect.

Uncertainty expression and reliance. Research into AI-assisted decision-making indicates that the manner in which AI expresses uncertainty can also influence people's reliance on it. In a preregistered study involving 404 participants, Kim et al. discovered that when uncertainty was expressed in the first person, it led to a decrease in agreement with the AI and enhanced accuracy [4]. Steyvers et al. found that longer explanations increased users' confidence without having any effect on accuracy, while matching the uncertainty expressed with the model's actual level of confidence improved calibration [5]. Buçinca et al. found that cognitive-forcing interventions decreased AI overreliance, though more intensive interventions were rated less favourably by the

users [7]. These results demonstrate that reliance on the AI depends not only on its accuracy but also on how its advice is presented.

Acceptance of incorrect AI-generated content. Students may still accept incorrect AI advice. Urban et al. found that ChatGPT assistance did not reduce the amount of incorrect AI-generated information included in students' writing [8]. Ren found that 62.4% of 342 undergraduates accepted incorrect AI advice with open ChatGPT access, compared with 39.7% after a brief metacognitive reflection prompt [9]. Although advice acceptance and advice shift are different measures, the 62.4% acceptance rate is substantially higher than the 20–30% shift commonly reported in advice-taking research [3]. This suggests that students can be highly receptive to incorrect AI advice and that this reliance can be affected by the interaction design.

Downstream effects. Bastani et al. found that GPT-4 improved students' performance during assisted practice, but students with unrestricted access later performed worse than those who had not used AI when working independently [6]. Thus, improved performance with AI assistance does not necessarily lead to improved independent performance.

The body of literature indicates that AI communication can have an effect on immediate reliance, while AI-assisted practice can influence subsequent independent performance. Yet these two areas have not been directly linked. The present study therefore looks into whether the level of certainty expressed by the AI during circuit-fault practice has an impact on students' later independent diagnostic performance.

3. Research Question and Hypotheses

The main research question is given below:

Does the level of certainty expressed by a generative-AI assistant during circuit-fault-diagnosis practice affect undergraduate EEE students' unaided diagnostic accuracy on previously unseen circuit faults two weeks later?

The following are the definitions of the key terms. A certainty expression is the presence or lack of epistemic hedging in AI recommendations that are otherwise identical in wording, the only difference between the two AI scenarios being their level of certainty. Unaided means that the individual has no access to an AI assistant, notes, or the internet during the delayed test. *Diagnostic accuracy* refers to the correct identification of the faulty component and fault mode using independently established ground truth. *Previously unseen* applies to circuits which were not shown during the practice phase.

Other secondary questions look at whether certainty has an effect on the harmful changes that occur when a correct initial diagnosis is changed to an incorrect one, and whether it influences the difference between the students' confidence and their actual accuracy at the later testing.

The idea is that the certainty shown by AI could have an effect on how credible it seems, which in turn might cause students to change their first diagnosis and could affect the depth with which they engage in reasoning about the circuit. This might then have an impact on their later independent diagnosis. While the study looks at the students' initial and final judgments, their confidence levels, and their performance when assessed later without assistance, it does not directly measure the amount of reasoning the students carried out during their practice.

H1. Students will be more likely to change a correct initial diagnosis to an incorrect one when the AI gives a high-certainty recommendation than when it gives a low-certainty recommendation.

H2. Two weeks after the practice session, students who received high-certainty AI recommendations will have lower diagnostic accuracy on the unaided test than students who received low-certainty recommendations or no AI assistance.

H3. The confidence–accuracy gap at delayed testing will be larger under high-certainty phrasing.

The study tests whether differences in AI certainty expression are associated with differences in subsequent independent performance. It is not intended to establish AI dependence as a general phenomenon across all forms of AI use or learning.

4. Proposed Methodology

4.1 Design, Participants and Study Arms

This study uses a three-arm randomized experiment with a two-week delayed post-test, since the research question calls for a causal comparison between conditions rather than a descriptive account of students' experiences with AI.

A total of 90 EEE undergraduate students who have completed introductory circuit analysis and electronic devices coursework will be recruited through an open invitation, independent of any course. A total of 110 students will be enrolled to allow for possible attrition. All participants will first complete a four-item unaided baseline test. They will then be block-randomized within baseline-score tertiles so that the three study arms are balanced according to prior ability. The no-AI control condition follows the design used in comparable AI-reliance research [7].

Arm	What the student sees after submitting an initial diagnosis
Control	A structured re-inspection step of matched duration, with no AI assistant.
High certainty	"The fault is an open R3. The node voltages are consistent with no current in that branch."
Low certainty	"The fault may be an open R3. The node voltages are consistent with no current in that branch."

The two AI conditions will be matched as closely as possible. The named component, fault mode, rationale, sentence count, word count, formatting and screen position will remain the same; only the level of hedging will change. This control is important because explanation length alone can increase user confidence without improving accuracy [5]. At the end of Session 1, participants will complete a manipulation check by rating how certain the AI assistant appeared on a scale from 0 to 100.

4.2 Items and Ground Truth

The study will use 22 single-fault items. They are four baseline items, ten practice items, and eight delayed-test items. Each item will present a circuit schematic together with a set of symptoms, such as node voltages, meter readings, or waveform captures. The symptoms will be generated by introducing one fault into an otherwise functioning circuit adapted from standard undergraduate material [10].

The faults will include open resistors, shorted capacitors, out-of-tolerance component values, reversed diodes, and failed amplifier stages. The correct diagnosis for each item will be established using three methods referring to hand analysis based on circuit equations, simulation

of the faulted netlist using [PSpice/LTspice / Multisim / Proteus], and independent review by two qualified assessors who are blind to the AI outputs. An item will be retained only when all three sources agree. The AI assistant will not be used to establish the correct answer.

4.3 AI Stimuli

One AI model and version will be used throughout the study with a fixed prompt and recorded temperature setting. The responses will be generated before participant enrolment and stored in advance, so participants will receive the same predetermined outputs rather than live AI responses. The model version, generation date, prompt, settings, and complete responses will be documented for later use, such as to support replication.

The practice set will contain seven correct and three incorrect AI recommendations, with the same recommendations presented in both AI conditions. These proportions will be specified before the responses are generated rather than selected afterward based on cases in which the model happens to make mistakes.

4.4 Procedure

13 Each practice trial will follow the same sequence: participants will give an initial diagnosis, rate their confidence from 0–100, view the assigned intervention, and then provide a final diagnosis and confidence rating. Participants in the AI conditions will also report their reliance on the AI. Requiring an initial diagnosis before viewing the AI recommendation allows changes in judgment to be measured directly.

Fourteen days after that, the participants will do eight diagnostic questions without any assistance and will time them as they go, then give their confidence level after each question. Four of the questions will involve the same fault categories as those used in the practice sessions on new circuits, and the other four will involve new circuit configurations. This arrangement separates out the element of retention from that of transfer, since the questions with the new circuit configurations test whether the students can apply their diagnostic ability to such new circuits rather than just recalling the patterns they have previously learned.

The participants are also going to report how many hours they spent on the circuit study, their use of AI, and the amount of course exposure they had during that period. These variables will be incorporated into the robustness analyses in order to see whether differences in exposure have an effect on the results.

4.5 Expected Result and Analysis

The main outcome is the accuracy with which students diagnose the eight items in Session 2, while the secondary outcomes are harmful reversals, beneficial revisions, appropriate resistance to AI advice, the confidence–accuracy gap, and the median diagnostic time. The confidence–accuracy gap is obtained by subtracting the percentage of correct answers from the mean confidence.

The analysis will employ a mixed-effects logistic regression in which the fixed effects are study condition, item type and their interaction, participant and item are the random effects and the baseline score acts as a covariate. There will be three pairwise comparisons: between the high and the low certainty groups, between the high certainty and the control group, and between the low certainty and the control group. Holm–Bonferroni correction will be applied to all three; the first two comparisons are confirmatory tests of H2, while the comparison between low certainty

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and control is exploratory. Hypothesis H1 will be examined using the harmful-reversal rates, H2 will be tested using the two confirmatory contrasts and H3 will be assessed via the confidence–accuracy gap.

The study seeks to have 30 fully enrolled participants in each group. Ren's effect is not taken into account when calculating power since the previous study looked at the immediate acceptance of wrong AI advice, while this one looks at delayed unaided accuracy. A sensitivity analysis will be carried out at the 0.05 alpha level and with 80% power in order to estimate the smallest detectable effect at both the unadjusted and the most stringent Holm-adjusted levels (with an alpha of 0.0167), and this will be reported as an odds ratio and as an absolute difference in accuracy.

4.6 Timeline and Ethics

During weeks 1-3, the study will focus on ethics approval and item construction. In Week 4, the items will be validated by experts, followed by a 12-participant pilot to assess item difficulty and the manipulation check. Session 1 will take place during weeks 5-6. Then it will be followed by Session 2 in Weeks 7-8. The remaining weeks 9-12 will be used for data analysis and writing.

Participants will be informed during consent that AI recommendations may contain errors and should be evaluated as they would be in normal use. The specific items containing incorrect recommendations will not be disclosed, since doing so would change the task into an exercise focused on identifying AI errors rather than diagnosing circuits.

Participation will be voluntary and will have no academic consequences. Participant responses will be pseudonymized and will not be shared with instructors. At the end of the study, all participants will receive worked solutions for every item so that the three study arms receive the same instructional content.

5. Significance

Bangladesh's University Grants Commission announced in April 2026 that it was formulating a national policy on the ethical and responsible use of AI in higher education [11]. This highlights the need for evidence on how AI should be used and communicated in education.

If H2 is supported, the results would be in agreement with the idea that definite phrasing contributes to the downstream effects mentioned in AI-assisted learning [6]. Since the phrasing can be altered without limiting access, it could represent a less disruptive kind of intervention. However, if H2 is not supported, definite phrasing might not have a detectable impact on delayed unaided accuracy and other factors such as reasoning offloading could then serve as possible explanations.

The study also includes an engineering task which has objectively verifiable answers; circuit equations and simulation make it possible to identify harmful and beneficial diagnostic changes without having to depend entirely on subjective judgment.

6. Limitations

There are some important limitations to the study. Confidence is based on ratings given by the participants themselves, and the experiment took place at a single institution, within one engineering field, using one version of the AI model. Consequently, the results may not be directly applicable to other institutions, different areas of study, or to various AI systems, particularly as newer versions of the models become available.

The two-week interval between the sessions can be measured but cannot be fully controlled, which is why information about participants' AI use, study time, and course exposure during this period will be collected and included in the analysis. In addition, the planned sample size is sufficient mainly for detecting moderate and large effects, so smaller effects may not be detected reliably.

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